

SE-LAB 1

Problem statement #59

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1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match newly logged Found Item posts with active Lost Item reports based on category, colour tags, and location radius.	High	Pass: Match probability is calculated and a notification is sent to the possible owner. Fail: Items from different categories are matched.	This is the main function of the app, so it has to work correctly.
FR-002	Functional	The system shall allow a Finder to create a Found Item post with a photo, category, colour tags, and the approximate location where the item was found.	High	Pass: Post is saved and shows up in the Found Items list with photo, category and tags filled in. Fail: Post is saved without a photo or without a category.	The matching engine needs these tags to compare found items with lost reports.
FR-003	Functional	The system shall allow an Owner to create a Lost Item report with a category, description, colour tags, and the approximate location where the item was lost.	High	Pass: Lost Item report is saved and becomes visible to the matching engine within 1 minute of submission. Fail: Report is saved but is not picked up for matching.	Owners need a simple way to register what they lost so the system can search for it.
FR-004	Functional	The system shall let an Owner view the list of potential matches for their Lost Item	Medium	Pass: Owner opens "My Lost Reports" and sees matched Found posts	Just getting one notification is not enough; the owner

		report, sorted by match probability.		sorted with the highest probability match first. Fail: Matches are shown in random order or without a probability score.	should be able to compare all possible matches.
FR-005	Functional	The system shall allow a Community Admin to remove a Found Item or Lost Item post that has been flagged as fake or suspicious.	Medium	Pass: Flagged post is removed from public listings and a removal log entry is created. Fail: Post is still visible after the admin removes it.	Keeps the platform trustworthy and stops people from misusing it.
NFR-001	Nonfunctional (Performance & Security)	Claimants must correctly answer the private security questions set by the Finder before the Finder's contact details are revealed.	High	Pass: Benchmarking tests confirm the target response time and security checks hold even under simulated peak load. Fail: Contact details are shown before the questions are answered correctly, or response time exceeds the target under load.	Protects the finder's personal contact details and confirms the claimant is the real owner.
NFR-002	Nonfunctional (Privacy & Security)	The system shall never show a user's exact address to another user; only an approximate distance (for example, "within 2 km") shall be shown.	Medium	Pass: A manual check of 10 random posts confirms that only an approximate distance is shown, never a full address. Fail: Any post shows an exact street address to another user.	Users should feel safe posting about a lost or found item without giving away where they live.

Use-Case Flow — View & Claim Matched Item

Use Case: View & Claim Matched Item (UC-04)

Actors: Owner (primary), Finder (receives the claim notification)

Preconditions:

1. Owner has an active Lost Item report in the system.
2. The system has already found at least one potential match for that report (UC-03 has run).
3. Owner is logged in to their account.

Postconditions:

1. If the claim succeeds, the Finder's contact details are shown to the Owner and the item is marked "Claim in progress."
2. If the claim fails, the item stays available and no contact details are shown.

Main Success Scenario:

1. Owner opens "My Lost Reports" and picks a report that has matches.
2. System shows the list of possible Found Item matches, highest match probability first.
3. Owner opens a match to see the full details (photo, category, tags, distance).
4. Owner taps "This is my item" to start the claim.
5. System shows the proof-of-ownership questions the Finder set for that item (this step includes UC-05: Verify Proof of Ownership).
6. Owner answers all the security questions.
7. System checks the answers against what the Finder saved.
8. If every answer is correct, the system marks the claim as verified and shows the Finder's contact details to the Owner.
9. System notifies the Finder that their found item has been claimed.
10. Use case ends successfully.

Alternate Flow:

7a. Incorrect Answer

7a1. System shows a "some answers are incorrect" message and does not reveal the contact details.

7a2. Owner can retry, up to 3 attempts in total.

7a3. If all 3 attempts fail, the system locks the claim on that item for 24 hours and tells the Owner to contact the Community Admin for manual verification.

Use Case Diagram



